

Combinatorial Theory: Positroids / 9.16.26

(Recall) we have four definitions for matroids.

↳ (elts $I \in \mathcal{M}$ are called bases of $\mathcal{M}$).

We have two axioms (exercise: show they are equiv.)

→ (Strong Exchange): For all $I, J \in \mathcal{M}$, and for all $i \in I$, there exists $j \in J$ such that

$$1) (I \setminus \{i\}) \cup \{j\} \in \mathcal{M},$$

$$2) (J \setminus \{j\}) \cup \{i\} \in \mathcal{M}.$$

→ (Weak Exchange): For all $I, J \in \mathcal{M}$, and for all $i \in I$, there exists $j \in J$ such that

$$(I \setminus \{i\}) \cup \{j\} \in \mathcal{M},$$

Def'n (Matroids - order-theoretic):

↳ partial order on $\binom{[n]}{k}$ such that

$$I = \{i_1 < i_2 < \dots < i_k\} \leq \{j_1 < j_2 < \dots < j_k\}$$

iff $i_1 < j_1, i_2 < j_2, \dots$ (called Gale order or "dominance order")

← for matroids

↙ not the one on partitions

For a permutation $w \in S_n$, $I \leq_w J$ if $w(I) \leq w(J)$

→ for all $w \in S_n$, $\mathcal{M}$ has a unique element under " $\leq_w$ "

↳ (called "Gale Axiom")

Thm: Strong/Weak Exchange axioms are equiv. to Gale axiom

↳ compare this with the fact that Matroid Stratification of $\text{Gr}_{k,n}$ is the common refinement of $n!$ Schubert decomp.

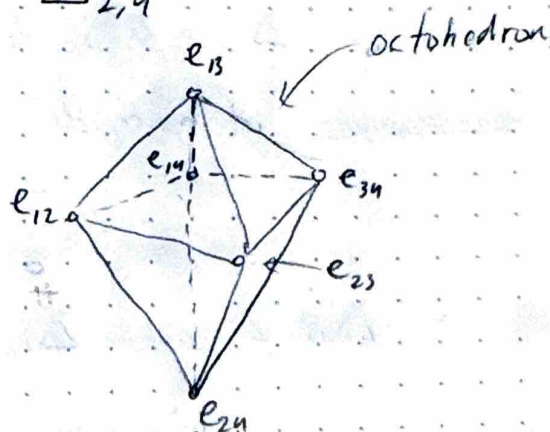
Let's now talk about polytopes...

Def'n (Hypersimplex):

$$\hookrightarrow \Delta_{k,n} = \text{Conv Hull} \left(\begin{array}{l} \text{binary vectors } e_I \\ \text{with } k \text{ 1's and } (n-k) \text{ 0's} \end{array} \right)$$

namely, $e_I := \sum_{i \in I} \vec{e}_i$

Ex consider $\Delta_{2,4}$



Note: all binary vectors e_I are vertices of $\Delta_{k,n}$

Note: all edges of $\Delta_{k,n}$ are parallel to $\vec{e}_i - e_j$

(i.e. the edges connect e_I and e_J connected by switching 1 and 0 in positions i and j .)

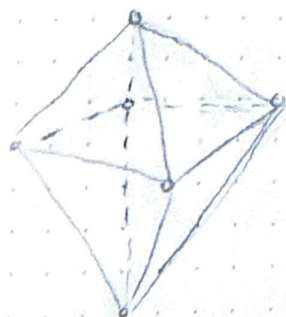
Now, consider $M \in \binom{[n]}{k}$ and let $P_M := \text{Conv}(e_I, I \in M)$

Thm (Gelfand-Goresky-McPherson-Serganova):

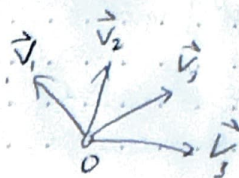
$\hookrightarrow M$ is a matroid iff any edge P_M is parallel to $(\vec{e}_i - \vec{e}_j)$

(Such P_M are called Matroid Polytopes)

EX: $k=2, n=4$ (uniform rank-4 matroid)



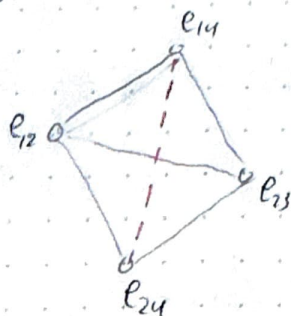
→ four basis vectors



$$\Delta_{24} = 0, \quad \vec{v}_2 \parallel \vec{v}_4$$

we also have some non-examples of matroids

EX:



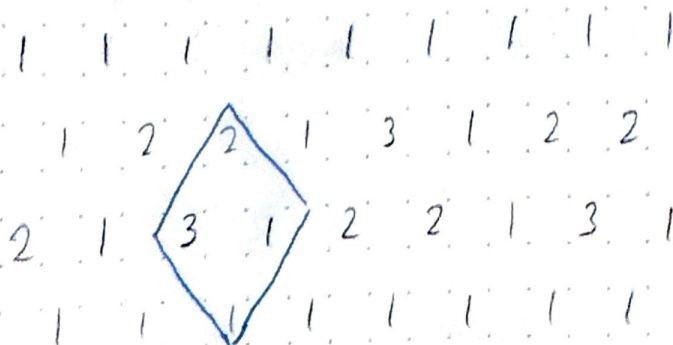
$$\begin{aligned} \overset{0}{\parallel} \Delta_{13} \cdot \overset{0}{\parallel} \Delta_{24} &= \overset{0}{\neq} \Delta_{12} \cdot \overset{0}{\parallel} \Delta_{34} \\ &+ \overset{0}{\parallel} \Delta_{14} \cdot \overset{0}{\neq} \Delta_{23} \end{aligned} \quad \left. \vphantom{\begin{aligned} \overset{0}{\parallel} \Delta_{13} \cdot \overset{0}{\parallel} \Delta_{24} &= \overset{0}{\neq} \Delta_{12} \cdot \overset{0}{\parallel} \Delta_{34} \\ &+ \overset{0}{\parallel} \Delta_{14} \cdot \overset{0}{\neq} \Delta_{23} \end{aligned}} \right\} \begin{array}{l} \text{3-term} \\ \text{Plücker} \\ \text{relation} \end{array}$$

Exchange axioms fail for $I = \{2, 4\}$
and $J = \{1, 3\}$, w/ $i=2$.

There are some matroids that don't come from collections of vectors, which are called non-realizable matroids.

Now, let's do something unrelated. Frieze Patterns

EX: a frieze pattern of order 5

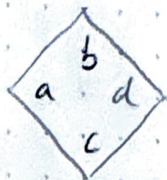


$$3 \cdot 1 - 2 \cdot 1 = 1$$

Def'n (Frieze Patterns of order n):

↳ infinite horizontal strips of integers with

- 1) $n-1$ total rows
- 2) top and bottom rows have all 1's
- 3) for all diamonds



we have $ad - bc = 1$.

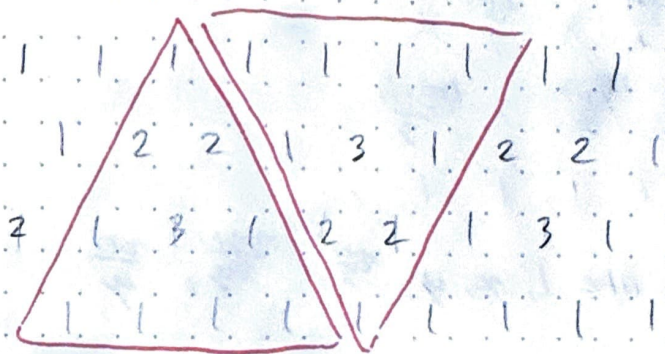
(secretly related to the 3-term Plücker relation)

Thm (Coxeter):

↳ all rows are periodic with period n .

Moreover, the following stronger "glide symmetry" property holds.

ex:



Thus, we get a general pattern



w/ $\tilde{T}$ obtained as the mirror image of T , shifted to the right.

How many frieze patterns of order n do we have?

→ $n=2$, just one

... | | | | | ...

→ $n=3$, just one

| | | | |
| | | | |

→ $n=4$, we have at least two

| | | | |
x $\frac{2}{x}$ x $\frac{2}{x}$
| | | | |

→ $n=5$, we have five...

| | | | |
x $\frac{y+1}{x}$ x $\frac{y+1}{x}$ x $\frac{y+1}{x}$
 $\frac{(y+1)(1+\frac{y}{x})}{y}$ $\frac{(y+1)(1+\frac{y}{x})}{y}$ $\frac{(y+1)(1+\frac{y}{x})}{y}$ $\frac{(y+1)(1+\frac{y}{x})}{y}$
| | | | |

terms will evolve like $1, x, y, \frac{y+1}{x}, \frac{xy+y+1}{xy}, \frac{xy+1}{y}$

Thm: the number of frieze patterns of order $n = C_{n-2} = \frac{1}{n-1} \binom{2n-4}{n-2}$

↳ C_{n-2} is the $n-2$ 'nd Catalan number.

Suppose $n = \infty$, we get

1	2
1	5
1	2
1	5
1	2
1	5
1	2
1	5
1	2
1	5
1	2
1	5

-- continues w/
every other fibonacci number